

Applications of operator algebras to PDEs

– layer potentials and boundary value problems –

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Based on works with **Marius Mitrea** (Baylor) and **Mirela Kohr** (Cluj)



Abstract and outline

- I will talk about the **method of layer potentials**, one of the oldest methods to solve boundary value problems (Gauss, ...).
- It is more complex than the “energy methods,” but presents *additional advantages* (including some practical ones: numerical methods, Dirichlet-to-Neumann maps (Grieser)).
- Historically, it was (to my knowledge) one of the main motivations for the development of operator algebras (compact and Fredholm operators, Hilbert spaces, ...).
- I will make the return trip of applying ideas from operator algebras to obtain new results on layer potentials on some **non-compact** domains (cyl. ends: “non-compact= singular,” Melrose.)
- *Pseudodifferential op. will constantly be in the background.*

Outline

- 1 Review of layer potentials for $\Omega \subset \mathbb{R}^n$ bounded, smooth
- 2 The Mitrea–Taylor trick: Fredholm operators and index theory
- 3 Manifolds with cylindrical ends and operator algebras

A large part of my talk is on *background material*.

The new results are mostly joint work with **Mirela Kohr**

However, some results are based on older results with **Catarina Carvalho**, **Richard B. Melrose**, **Marius Mitrea**, **Yu Qiao**, and **Wolfgang L. Wendland**

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Boundary value problems and layer potentials

Consider the *classical Dirichlet problem*

$$(D) \quad \begin{cases} \Delta u = 0 & \text{in } \Omega \\ u = f & \text{on } \partial\Omega. \end{cases}$$

The **method of layer potentials** reduces this problem to one of the following two **pseudodifferential equations** on $\partial\Omega$:

- $\mathbf{S}g_1 = f$ (*single layer potential operator*) or
- $(\frac{1}{2} + \mathbf{K})g_2 = f$ (*double layer layer potential operator*) on $\partial\Omega$,

My talk:

I will explain how these reductions are obtained and how to solve the pseudodifferential equations on *domains with cylindrical ends*.

The definition of single and double layer potentials

The *single* and *double layer potentials* $\mathcal{S}$ and $\mathcal{D}$ are defined as $\Delta^{-1}\phi$ for particular distributions ϕ **supported** on $\partial\Omega$.

Let us consider the distribution $\phi := g \otimes \delta_{\partial\Omega}$ defined by

$$\langle g \otimes \delta_{\partial\Omega}, h \rangle := \int_{\partial\Omega} g(x)h(x)d\sigma(x)$$

($d\sigma$ is the **surface measure** on $\partial\Omega$). Then

$$\mathcal{S}g(x) := \Delta^{-1}(g \otimes \delta_{\partial\Omega}).$$

and

$$\mathcal{D}g(x) := \Delta^{-1}[\partial_{\nu}^*(g \otimes \delta_{\partial\Omega})],$$

where ν is the *outer unit normal vector field* to $\partial\Omega$.

The single and double layer potentials are harmonic

The formula $\mathcal{S}g(x) = \Delta^{-1}(g \otimes \delta_{\partial\Omega})$ just discussed also gives

$$\Delta \mathcal{S}g(x) := \underbrace{\Delta[\Delta^{-1}(g \otimes \delta_{\partial\Omega})]}_{id}(x) = g \otimes \delta_{\partial\Omega}(x) = 0,$$

for $x \in \Omega$, since $g \otimes \delta_{\partial\Omega}$ is **supported** on $\partial\Omega$.

So the simple layer potentials are **harmonic outside** $\partial\Omega$.

A similar proof gives that $\Delta \mathcal{D}g = 0$ outside $\partial\Omega$, using this time

$$\mathcal{D}g = \Delta^{-1}[\partial_\nu^*(g \otimes \delta_{\partial\Omega})].$$

(Recall that ν is the *outer unit normal* vector to $\partial\Omega$.)

The classical formulas

Recall that

$$\Delta^{-1}g(x) = \int_{\mathbb{R}^n} \frac{c_n}{|x-y|^{n-2}} g(y) dy, \quad (n \geq 3).$$

This formula and the definitions of $\mathcal{S}$ and $\mathcal{D}$ give (classical)

$$\mathcal{S}g(x) := c_n \int_{\partial\Omega} |x-y|^{2-n} g(y) d\sigma(y) \text{ and}$$

$$\mathcal{D}g(x) := c_n \int_{\partial\Omega} \partial_{\nu_y} |x-y|^{2-n} g(y) d\sigma(y).$$

By derivation under the $\int$ sign, we again obtain that $\mathcal{S}g$ and $\mathcal{D}g$ are **harmonic outside** $\partial\Omega$.

The method of layer potentials

To **summarize**: the single and double layer potentials $\mathcal{S}g$ and $\mathcal{D}g$ are harmonic functions *outside* Ω . So **they satisfy the first equation** of our Dirichlet problem

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u = g & \text{on } \partial\Omega. \end{cases}$$

The method of layer potentials is then to look for u of the form

$$u = \mathcal{S}g \quad \text{or} \quad u = \mathcal{D}g.$$

What about the *boundary conditions*?

Answer: **pseudodifferential operators** (more in Mirela Kohr's talk).

Boundary conditions

Consider the boundary conditions (i.e., the limit)

$$\mathbf{S}g(z) := \lim_{x \rightarrow z} \mathcal{S}g(x), \quad z \in \partial\Omega.$$

“Same” formula because we can pass to the limit under the $\int$:

$$\mathbf{S}g(z) := c_n \int_{\partial\Omega} |z - y|^{2-n} g(y) d\sigma(y), \quad z \in \partial\Omega.$$

Then $\mathbf{S}$ is a **pseudodifferential** operator of order -1 and we have

$$\mathcal{S}(g)|_{\partial\Omega} = \mathbf{S}g.$$

So to solve our Dirichlet problem (D), **it is enough to invert $\mathbf{S}$** .

As we will see, this involves a significant amount of **operator theory**, (the rest of my talk).

Double layer potential

We prefer the analogous **double layer potential** operator $\mathcal{D}$, namely:

$$\mathcal{D}g(x) := \Delta^{-1}[g \otimes \delta'_{\partial\Omega}] = c_n \int_{\partial\Omega} \partial_{\nu_y} |x - y|^{2-n} g(y) d\sigma(y).$$

Difference: for $x \rightarrow z \in \partial\Omega$, we obtain additionally (“jump relation”):

$$\begin{aligned} \lim_{x \rightarrow z} \mathcal{D}g(x) &= c_n \text{p.v.} - \int_{\partial\Omega} \partial_{\nu_y} |z - y|^{2-n} g(y) d\sigma(y) + \frac{g(z)}{2} \\ &= \left(\mathbf{K} + \frac{1}{2} \right) h(z), \quad z \in \partial\Omega. \end{aligned}$$

Summary of the double layer potential method

Our Dirichlet problem reduces to solving the (pseudodiff.) equation

$$\left(\frac{1}{2} + \mathbf{K} \right) h = f.$$

The case of smooth domains

Basic fact

An order -1 pseudodifferential operator on a *compact, smooth manifold* is a **compact** operator.

Recall that (up to a factor) **K** is the integral operator with kernel

$$\partial_{\nu_y} |z - y|^{2-n} = (2-n) \frac{(z - y) \cdot \nu_y}{|z - y|^n}.$$

If $\partial\Omega$ is **smooth**, then **K** is an **order -1 pseudodifferential op.**

If Ω is also bounded, then **K** is **compact**.

Definition

Let $T : X \rightarrow Y$ be a continuous, linear map between *Banach spaces*. We say that T is **Fredholm** if $T^{-1}(0)$ and Y/TX are *finite dimensional*.

Solvability for Ω smooth, bounded

We can now prove the Hadamard well posedness (solvability) of our Dirichlet problem (D) if Ω is **smooth and bounded**:

- Since $\mathbf{K}$ is of order -1 , we know that it is **compact**, and hence $\mathbf{P} := \frac{1}{2} + \mathbf{K}$ is **Fredholm of index** $:= \dim T^{-1}0 - \dim T^{*-1}0 = 0$.
- Energy estimates and the restriction to the boundary (or “jump”) relations show that $\mathbf{P}$ has **zero kernel**.
- Hence $\mathbf{P}$ is **invertible** and the solution to our Dirichlet problem is

$$u(x) := \mathcal{D}(\mathbf{P}^{-1}f) = \mathcal{D}\left(\left(\frac{1}{2} + \mathbf{K}\right)^{-1}f\right).$$

Indeed, we already noticed that u is *harmonic*. Then

$$u|_{\partial\Omega} := \mathcal{D}\left(\left(\frac{1}{2} + \mathbf{K}\right)^{-1}f\right)|_{\partial\Omega} = \left(\frac{1}{2} + \mathbf{K}\right)\left(\frac{1}{2} + \mathbf{K}\right)^{-1}f = f.$$

Extensions

This completes our discussion of layer potentials on a **smooth, bounded domain** Ω .

If Ω is not smooth *or* not bounded, **K** may **no longer be compact**.
(Observed in physical phenomena.)

We want to extend these results to the case when Ω is a **domain with cylindrical ends**. (A more complicated **operator algebra**.)

Extensions to manifolds: *M. Mitrea and M. Taylor*, building on work by Fabes, Jerison, Kenig, Pipher, Riviere, Verchota, ...

The Mitrea–Taylor “trick”

Let us assume that $\Omega \subset M$ is a smooth manifold.

The first issue: $\Delta := d^*d$ may no longer be invertible on M .

The **Mitrea–Taylor “trick”** consists of the following:

- ① to replace Δ with $\Delta + V$, where V = potential,
- ② to replace Δ^{-1} (which may not exist) with $(\Delta + V)^{-1}$, and
- ③ to set

$$\mathcal{S}g(x) := (\Delta + V)^{-1}(g \otimes \delta_{\partial\Omega}) \quad \text{and}$$

$$\mathcal{D}g(x) := (\Delta + V)^{-1}[\partial_\nu^*(g \otimes \delta_{\partial\Omega})].$$

Hence we need to invert $\Delta + V$, (then we will invert $\frac{1}{2} + \mathbf{K}$).

Why does the “Mitrea-Taylor trick” work?

Let $V \in \mathcal{C}^\infty(M)$ satisfy:

- $V \geq 0$ on M and $V = 0$ on Ω ,
- V is not identically 0 on M ($V \not\equiv 0$ on M).

Assume M to be **smooth and compact**, then:

- $\Delta + V \geq 0$ is **Fredholm of index zero**.
- If $(\Delta + V)u = 0$, then

$$0 = ((\Delta + V)u, u) = (du, du) + (Vu, u),$$

and hence $Vu = 0$ and $du = 0$. Since $V \not\equiv 0$, $u = 0$. We obtain that $\Delta + V$ is **injective**.

- Consequently, $\Delta + V$ is **invertible** and **we can define $\mathcal{S}$ and $\mathcal{D}$ and proceed as before** to prove well-posedness. (With M. Kohr, we have extended this to **manifolds with cylindrical ends**.)

“Pictorial” definition of **cylindrical ends**

$\Omega = N$ has a boundary and M doesn't (both have straight cyl. ends)

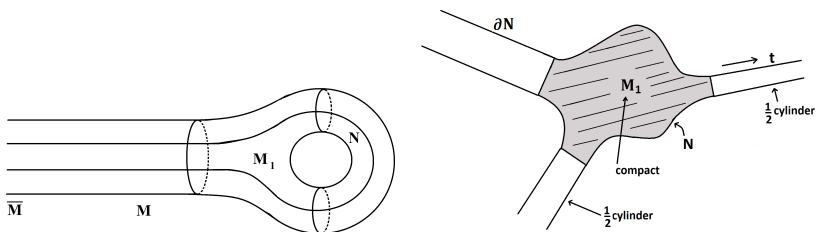
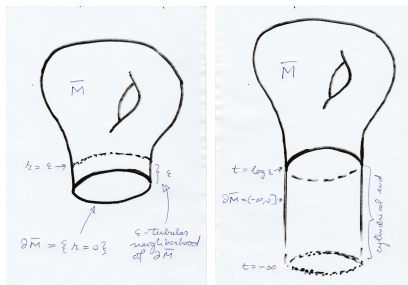


Figure: Manifolds with boundary (or domains) with cylindrical end

The operator **K** is **not compact** anymore (related to physical phenomena! Putinar, ...).

Also related to **wave-guides**: Bonnet-Ben Dhia, Bonnetier, Chesnel, Niclas, Krejcirik, ...

Formal definition of M with cylindrical ends



- $\bar{M}$ = a smooth, compact Riemannian manifold, $\partial \bar{M} \neq \emptyset$.
- To $\bar{M}$ we attach the cylindrical end $\partial \bar{M} \times (-\infty, 0]$ (prod. metric)

$$M := \bar{M} \cup \partial \bar{M} \times (-\infty, 0]$$

(A particular case of *Lie manifolds*, Ammann-Lauter-V.N.)

Main result

Theorem (Mitrea-V.N., Kohr-V.N.)

Let $\Omega \subset M$ be a domain with straight cylindrical ends and let $V \geq 0$ be translation invariant and non-zero at infinity. Then we have

- $\Delta + V$ is **invertible**, and hence the double layer potential operator K associated to the $\Delta + V$ is defined.
- $\frac{1}{2} + K$ is also **invertible**, and hence, if $V = 0$ on Ω , then the Dirichlet problem on Ω is well posed.

We need to consider **suitable classes of (pseudodifferential) operators** that take into account the (translation invariant) structure of the Laplacian on manifolds with cylindrical ends. Our choice: the smallest algebra with these properties (inspired by an unpublished work with Melrose).

Translation invariant operators near infinity

Consider the **partial translations** (isometries)

$$\Phi_s(x, t) := (x, t - s), \quad s \geq 0$$

$$\Phi_s : \partial \overline{M} \times (-\infty, 0] \rightarrow \partial \overline{M} \times (-\infty, -s].$$

Definition

$P : C_c^\infty(M) \rightarrow C_c^\infty(M)$ is called **translation invariant at infinity** if

- its Schwartz kernel is supported in an ε -neighborhood of the diagonal (so properly supported) and
- there is $R > 0$ s.t. if $\text{supp}(f) \subset \partial \overline{M} \times (-\infty, -R)$ and $s > 0$,

$$P\Phi_s(f) = \Phi_s P(f).$$

Equivalent to $P(x, t, x', t') = P(x, t-s, x', t'-s)$ for $s > 0$ and $-t, -t'$ large (in distribution sense).

Spaces of translation invariant op. near infinity

Definition

Let $\Psi_{\text{inv}}^m(M)$ be the space of classical pseudodifferential operators P of order $\leq m$ that are **translation invariant in a neighborhood of infinity**, that is

$$P(t, x, t', x') = P(t - s, x, t' - s, x'),$$

if $-t, -t'$ are large enough and $s \geq 0$.

These definitions extend right away to operators acting on vector bundles.

(Not a “groupoid” pseudodifferential algebra!)

Almost translation invariant operators at infinity

Drawback: $\Psi_{\text{inv}}^{\infty}(M)$ is not stable under inversion/holomorphic functional calculus.

Definition

$\Psi_{\text{ess}}^{\infty}(M)$ = *operators essentially translation invariant:*

- $\Psi_{\text{ess}}^m(M) := \Psi_{\text{inv}}^m(M) + \Psi_{\text{ess}}^{-\infty}(M)$
- $\Psi_{\text{ess}}^{-\infty}(M)$ = **suitable closure** of $\Psi_{\text{inv}}^{-\infty}(M)$ (w.r.t the norms $\|\cdot\|_{m',m}$ of linear operators $H^{m'}(M) \rightarrow H^m(M)$, $m', m \in 2\mathbb{Z}$ and with respect to the norms $\|ad_x^k(P)\|_{m',m\cdot}$)

Operator algebras: stable under holomorphic functional calc.

$\Psi_{\text{ess}}^{\infty}(M)$ is larger than $\Psi_{\text{inv}}^{\infty}(M)$ and stable under hol. funct. calc.

Spectral invariance

- $\Psi_{\text{inv}}^{\infty}(M)$ is a subset of the **b -calculus** studied by Melrose (also by C. Anghel-Stan, Grieser, Lesch, Schulze, and many others).
- However, $\Psi_{\text{ess}}^{\infty}(M)$ is not a subset of the b -calculus (at least if considering the Kondratiev compactification). Nevertheless, the understanding of the b -calculus is crucial.
- It is a (strict) subset of the c -calculus (Grieser, Mazzeo, Melrose).

Theorem (Kohr-Mitrea-V.N.)

Let $T \in \Psi_{\text{ess}}^m(M)$ $m \geq 0$, be such that T is **invertible** as a (possibly unbounded operator on $L^2(M)$). If $m > 0$, we assume also that T is **elliptic**. Then $T^{-1} \in \Psi_{\text{ess}}^{-m}(M)$.

The need of spectral invariance

The proof is based on the same ideas as the ones in Zenobi's talk :-)

The **spectral invariance** of $\Psi_{\text{inv}}^{\infty}(M)$ is needed in order to define the layer potentials:

$$\mathcal{S}g(x) := (\Delta + V)^{-1}(g \otimes \delta_{\partial\Omega}) \quad \text{and}$$

$$\mathcal{D}g(x) := (\Delta + V)^{-1}[\partial_{\nu}^{*}(g \otimes \delta_{\partial\Omega})].$$

For this, we need $\Delta + V$ to be **invertible**, hence $V \not\equiv 0$ is needed since Δ is not Fredholm on manifolds with cylindrical ends.

We also need **Fredholm conditions** (next).

Specific feature: **limit operator**

Definition

Let $P \in \Psi_{\text{ess}}^m(M)$. The **limit operator** associated to P is the unique **$\mathbb{R}$ -invariant** (i.e. translation invariant) order $\leq m$ pseudodifferential operator $\tilde{P} : C_c^\infty(\partial\overline{M} \times \mathbb{R}) \rightarrow C_c^\infty(\partial\overline{M} \times \mathbb{R})$ s.t.

$$\tilde{P}(t, x, t', x') := \lim_{s \rightarrow \infty} P(t - s, x, t' - s, x').$$

- $\tilde{P}$ is **well defined** and **translation invariant**.
- $\tilde{P}(t, x, t', x') := P(t - s, x, t' - s, x')$ if $P \in \Psi_{\text{inv}}^\infty(M)$.

(For more complicated geometries: **groupoids**: Androulidakis, Bohlen, Connes, Debord, Lescure, Mohsen, Skandalis, Yuncken, V.N.-Weinstein-Xu, ... , or compactifications: Anghel-Stan, Grieser, Kottke, Mazzeo, Melrose, Moroianu, Rochon,)

Next theorem: **a generalization of a classical result:**

Theorem (Kondratiev, Melrose and Mendoza)

Let $M = \overline{M} \cup (\partial \overline{M} \times (-\infty, 0])$ be a manifold with cylindrical and $P \in \Psi_{\text{ess}}^m(M; E, F)$. Then $P : H^s(M; E) \rightarrow H^{s-m}(M; F)$ is a **Fredholm operator** if, and only if,

- it is **elliptic** and
- its **limit operator** $\tilde{P} : H^s(\partial \overline{M} \times \mathbb{R}; E) \rightarrow H^{s-m}(\partial \overline{M} \times \mathbb{R}; F)$ is **invertible**.

“Best proof:” using **operator theory/operator algebras**.

More complicated spaces: **groupoids and primitive ideals** (V.N.-Prudhon; also: Georgescu, Măntoiu, Beltiță, ...).

Kohr–V.N.: for DN-elliptic operators,

Proof of the main result

The proof of our main result (the invertibility of $\Delta + V$ and of $\frac{1}{2} + K$) is done then as in the compact case (outlined earlier for Δ):

- The Fredholm property theorem (just mentioned) implies that $\Delta + V$ is **Fredholm of index zero**.
- The properties of $\psi_{\text{ess}}^\infty(M) \ni (\Delta + V)^{-1}$ (Kohr-V.N.), including the jump relations, show that $\Delta + V$ is injective, and hence invertible. So the **layer potentials are defined**.
- The behavior near the boundary of the layer potentials (“jump relations”) shows that (Kohr-V.N.)

$$S, K \in \psi_{\text{ess}}^{-1}(M).$$

- In turn, we prove that $\frac{1}{2} + K$ is also Fredholm, then that it is of index zero, then that it is injective, and then we conclude that it is invertible. **Hence the well-posedness.**

Conclusion

Although the double layer potential operator $\mathbf{K}$ is **no longer compact** on a manifold with cylindrical ends, we still obtain the invertibility of $\frac{1}{2} + \mathbf{K}$ and hence **the well-posedness of the Dirichlet problem** (also Ammann-Große-V.N. for bounded geometry).

The role of **operator algebras**:

- ① Fredholm conditions.
- ② Index theory arguments (index zero).
- ③ Spectral invariance (P in the algebra $\Rightarrow P^{-1}$ in the algebra.)

Thank you for your attention!

Vielen Dank für die Aufmerksamkeit!